BIOCHEMISTRY CUMULATIVE EXAM

JANUARY 18, 1992

Consider the peptide:

Ala-Ser-Glu-Gly-His-Thr-Glu-Leu-Glu-Ile-Tyr-Phe-Asn-Gln

1. (50) Calculate the isoelectric pH of the peptide, assuming that the pK's of all the ionizable groups are the same as the standard values given in the table. Also calculate the fractional charge on each ionizable group at the pI. Show your work.

2. (30) Calculate how many ml of 1M NaOH it would take to titrate 1 liter of 0.1 M solution of the peptide from its isoelectric pH to pH 6.5.

3. (20) Calculate the net charge of the peptide at pH 6.5.

Table 2-4
pK values of some amino acids

Amino acid	*pK values (25°C)*		
	α-COOH group	*α-NH_3^+ group*	*Side chain*
Alanine	2.3	9.9	
Glycine	2.4	9.8	
Phenylalanine	1.8	9.1	
Serine	2.1	9.2	
Valine	2.3	9.6	
Aspartic acid	2.0	10.0	3.9
Glutamic acid	2.2	9.7	4.3
Histidine	1.8	9.2	6.0
Cysteine	1.8	10.8	8.3
Tyrosine	2.2	9.1	10.9
Lysine	2.2	9.2	10.8
Arginine	1.8	9.0	12.5

Source: After J. T. Edsall and J. Wyman. *Biophysical Chemistry* (Academic Press, 1958), ch. 8.